



School of Information Technology and Engineering
Discipline: Automated Control System Based on Simatic S7-1200

**Industrial Automation of a Final Assembly and Packaging System using
Siemens SIMATIC S7-1200**

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Almaty 2025,

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INTRODUCTION

Industrial automation has become a fundamental component of modern manufacturing, enabling higher productivity, improved product quality, and enhanced operational safety. With increasing market demand and growing product customization, traditional manual and semi-automated assembly and packaging processes are no longer sufficient. Such approaches often result in longer cycle times, increased human error, limited scalability, and reduced process traceability. As a result, industries are actively transitioning toward integrated automation solutions that combine programmable logic controllers, robotic handling systems, and human-machine interfaces into unified control architectures.

This project is dedicated to the development and simulation of an Automated Final Module Assembly and Packaging System based on a Siemens S7-1200 PLC. The selected process represents a typical discrete manufacturing scenario found in electronics and mechatronic production, where multiple processing stages must operate in a coordinated and reliable manner. The system is designed as a dual-stream production line, in which two identical modules are processed in parallel. Each module undergoes automated transportation, robotic handling, and secondary processing before both streams are synchronized at a central assembly station to form a single integrated product.

After assembly, the finished unit is transferred to a packaging station, where it is enclosed, sealed, and prepared for final output. Special attention is given to synchronization logic, interlocks, and fault handling to ensure safe and efficient operation of the entire system. The automation solution supports both automatic and manual operating modes and provides operator supervision through an intuitive HMI.

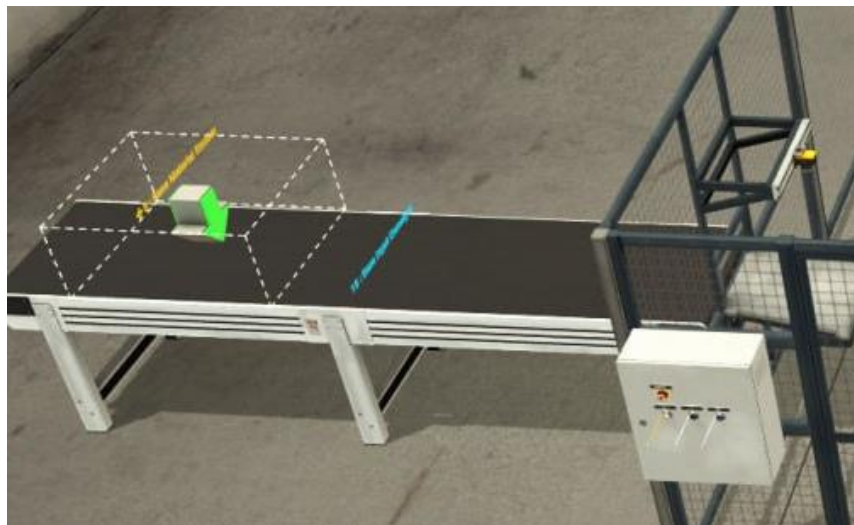
The scope of the project includes technological process analysis, development of a logical technological scheme, PLC programming, HMI visualization, and software-based simulation of system behavior. Mechanical design and physical commissioning are not considered. The primary goal is to demonstrate structured control logic, modular PLC design, and reliable coordination of parallel production processes in accordance with modern industrial automation practices.

1. Description of the technological process

The selected technological process represents a dual-stream automated final assembly and packaging line for electronic modules. Two identical production streams operate in parallel to increase throughput and reduce cycle time variability. Each stream processes one semi-finished module, which is later merged into a single final product.

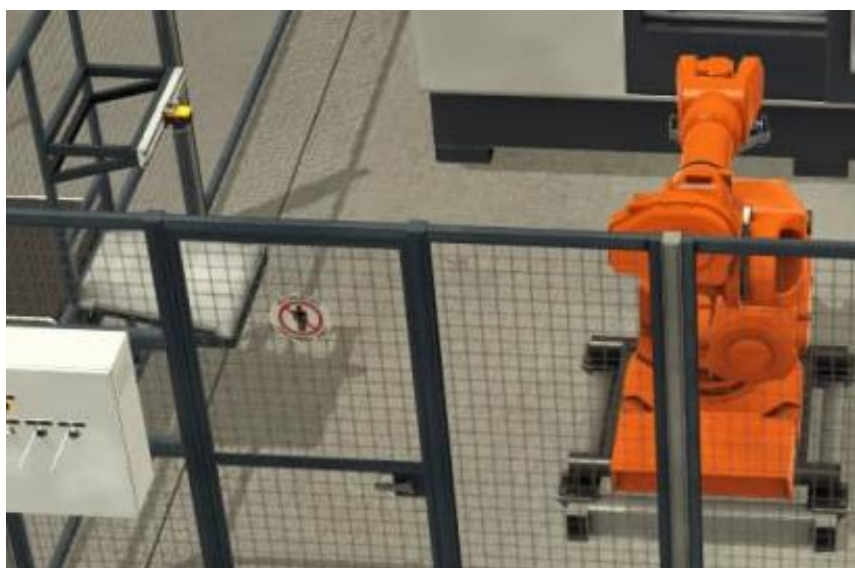
Module Infeed

Individual modules arrive on independent conveyor lines (Line A and Line B). Presence sensors detect module arrival and authorize further actions.



Robotic Pick-and-Place

An industrial robotic manipulator picks the module from the conveyor and places it into a dedicated Machining Center. The robot motion is synchronized with conveyor stop positions and safety interlocks.



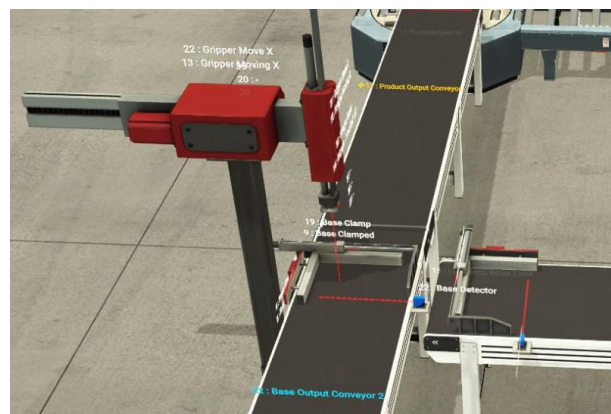
Machine Processing

Inside each Machining Center, a defined technological operation is executed. Depending on the application, this may include various operations



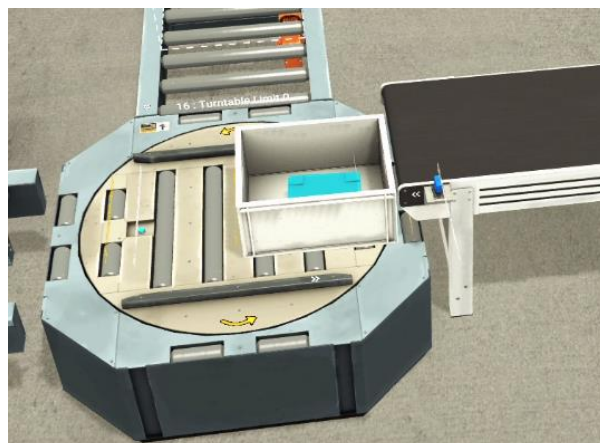
Central Assembly Operation

After both modules complete machining, they are transferred to a central assembly station. The two processed modules are aligned and joined into a single integrated unit using controlled actuators.



Packaging Process

The final assembled product is conveyed to a packaging station, where it is placed into protective packaging, sealed, and prepared for output.



1.1 Review of Modern Automation, Measurement, and Control Methods

Modern industrial automation systems increasingly rely on PLC-based distributed control, with Siemens S7-1200 widely used for small and medium-scale automated lines.

Structured sequencing (state machines) is recognized as best practices in recent industrial automation literature.

Sequential Programming

Most machine operations are sequential in nature, so it is necessary for the PLC to switch outputs depending not only on the input combinations but also on the current stage in the sequence. An output operating at the wrong time could cause damage or injury, so the correct programming technique is critical.

Recent studies indicate that modern assembly and packaging systems increasingly rely on PLC-based sequential control with distributed sensors and robotic handling. Parallel processing lines are commonly used to improve throughput and system flexibility.

Parallel Branching

It is sometimes desirable in sequential systems to carry out certain operations simultaneously. This has the advantage of saving time during each cycle. The SF chart technique known as parallel branching allows for some states to be active simultaneously and so the events run in parallel.

Human–Machine Interfaces provide operators with real-time monitoring, alarm handling, mode control, which is essential for safe industrial operation.

STEP 7 provides a user-friendly environment to develop, edit, and monitor the logic needed to control your application, including the tools for managing and configuring all of the devices in your project, such as controllers and HMI devices. To help you find the information you need, STEP 7 provides an extensive online help system.

1.2 Technological Parameters for Automation Design

- Conveyor speed: 0.2–0.5 m/s
- Pick-and-place cycle time: 18–20 seconds
- Start signal: Digital input from PLC
- Completion signal: Digital output to PLC
- Fault signal: Digital alarm output
- Assembly cycle time: 12–15 seconds
- Synchronization requirement: Both Line A and Line B must be complete
- Packaging cycle time: 8–10 seconds

- Package presence sensor: Digital
- Output conveyor speed: 0.3 m/s
- Emergency stop response time: < 100 ms

1.3 Industrial Automation Problem Statement

The objective is to design and implement a PLC-based industrial automation system using a Siemens S7-1200 controller that ensures safe, synchronized, and efficient operation of a dual-stream automated assembly and packaging line.

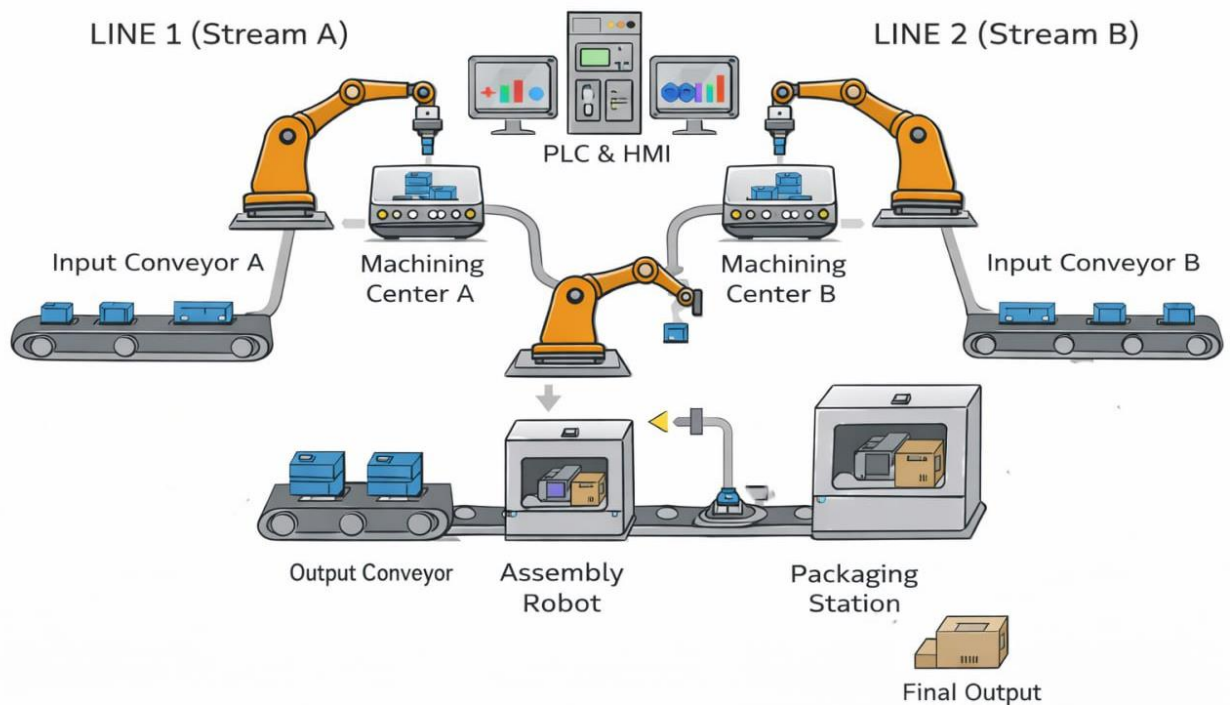
The automation system must:

- Control two independent production lines operating in parallel
- Coordinate robotic handling, machining operations, assembly, and packaging
- Synchronize both lines at a central assembly station
- Provide reliable interlocks and fault handling
- Supporting automatic and manual operating modes
- Enable operator monitoring and control through an HMI
- Allow software-based simulation and testing without physical hardware

2. TIA Portal Implementation & PLC Programming

The control logic for the Automated Final Module Assembly & Packaging System was developed using Siemens TIA Portal (Totally Integrated Automation Portal) V19. The PLC program was implemented using Structured Control Language (SCL) and Ladder Logic (LAD), enabling modular, readable, and maintainable code. Key functions—including conveyor control, robotic sequencing, sensor feedback handling, interlocks, and mode selection (automatic/manual)—were structured into networks with clear naming conventions and memory bit management. Timers, set/reset coils, and safety conditions (e.g., emergency stop, start authorization) were integrated to ensure reliable and safe operation. The TIA Portal environment facilitated seamless simulation, online monitoring, and HMI-PLC integration, allowing full validation of the system logic without physical hardware.

2.1 Technological Scheme Development



Process Units:

8 x Conveyors

Used for material transport between stations. Operate in indexed mode and controlled by digital outputs.

3 x Robotic Manipulators

Perform pick-and-place operations. Controlled via PLC start/stop signals with position and status feedback.

2 x Machining Center

Execute secondary processing operations. Operate as black-box units with start, busy, and done signals.

Position Sensor

Provide position detection of the controlled object

Presence sensor

Provide the presence of nearby objects to the control process

Table 1 Integration of Process Variables

Process Variable	Description	Application
Conveyor speed	m/s	Material transport

Position	Bool	Robot and actuator feedback
Processing time	seconds	Machining, assembly, packaging
Presence	Bool	Module detection
Seal status	Bool	Packaging quality confirmation

2.2 PLC Program Structure

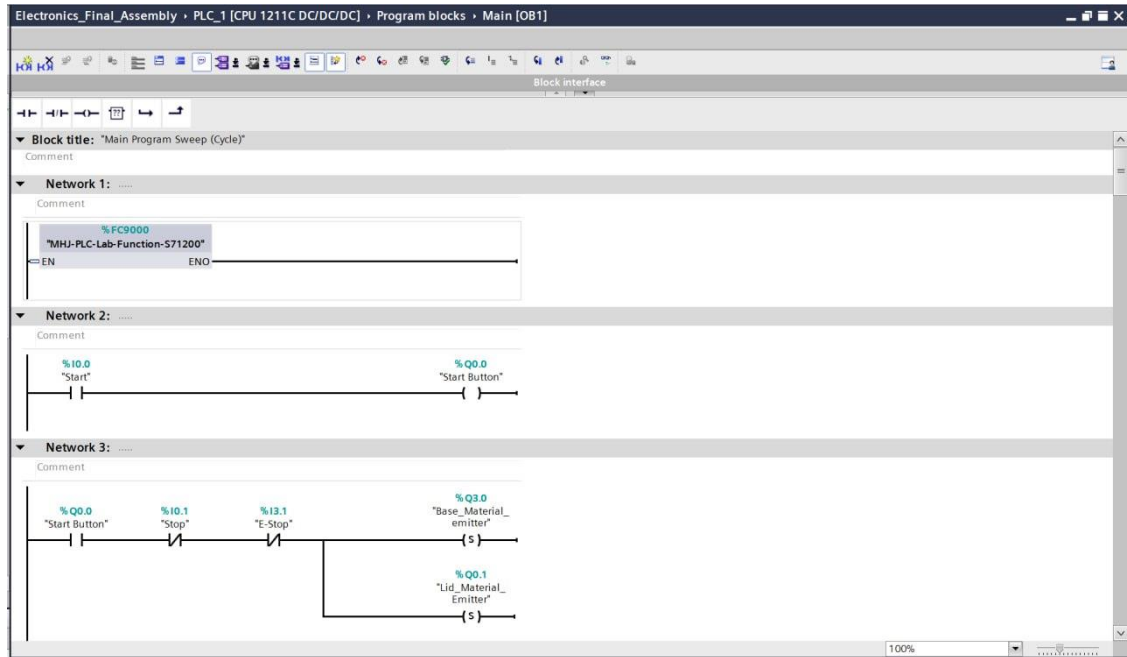


Figure 1 – Material Feed Activation

This segment handles the system-wide function calls and establishes the initial start conditions that trigger the "Set" coils for both the base and lid material emitters, provided no stop or emergency-stop conditions are active.

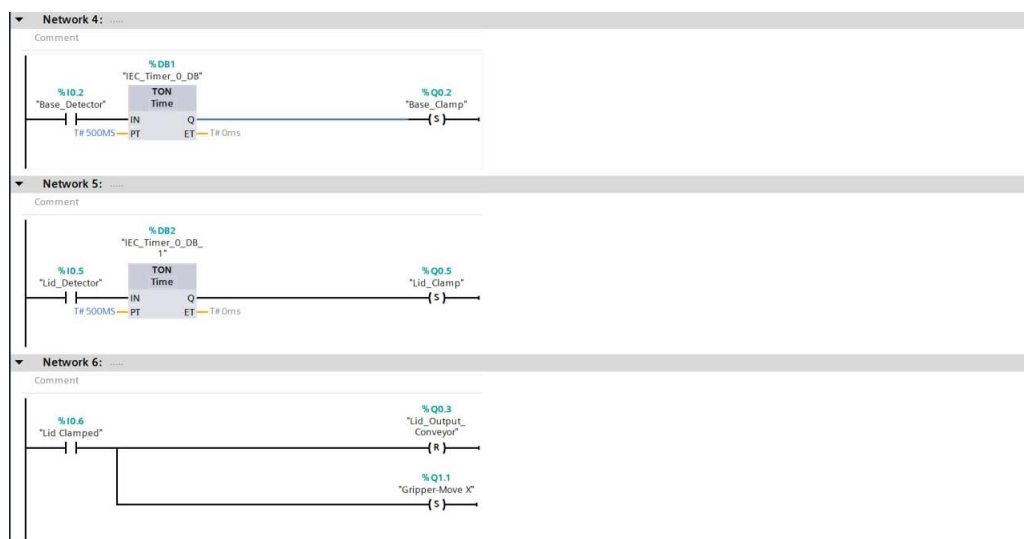


Figure 2 – Material Detection and Clamping

This logic employs 500ms on-delay timers to verify part presence via the base and lid detectors before activating the clamps and subsequently halts the lid conveyor to initiate the gripper's horizontal movement once the lid is secured.

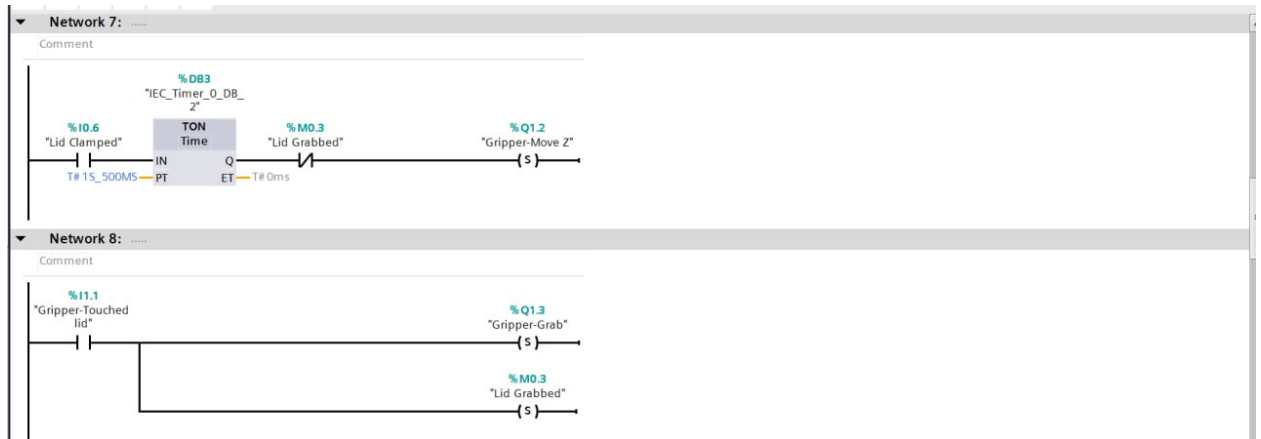


Figure 3 – Gripper Vertical Control and Material Handling

This logic utilizes an on-delay timer to trigger the gripper's Z-axis downward movement once the lid is clamped and secures the part by activating both the physical grab command and a "Lid Grabbed" memory bit upon contact.

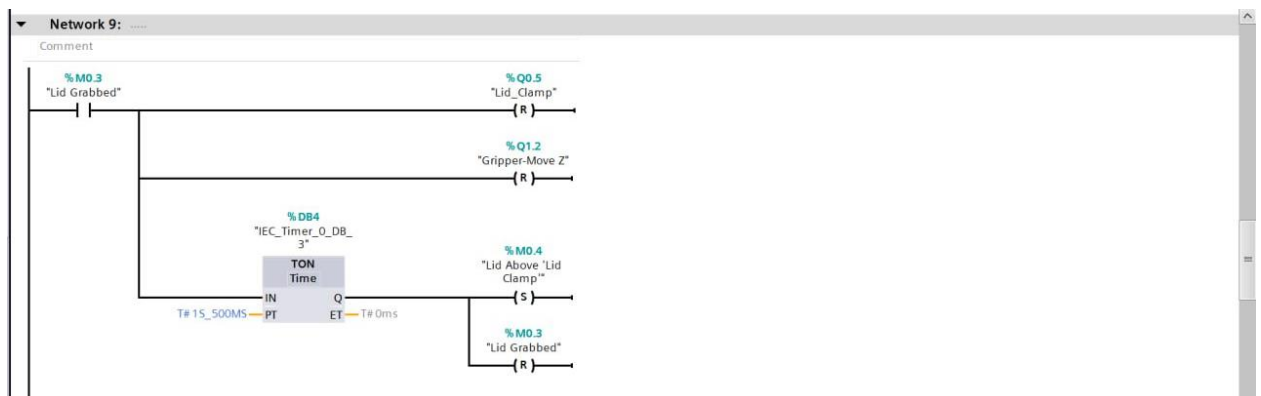


Figure 4 – Part Release and Clamping Sequence

This logic utilizes the "Lid Grabbed" status to release the lid clamp and reset the vertical gripper movement, while an on-delay timer eventually triggers a memory bit indicating the lid is clear of the clamping station and resets the grab status.

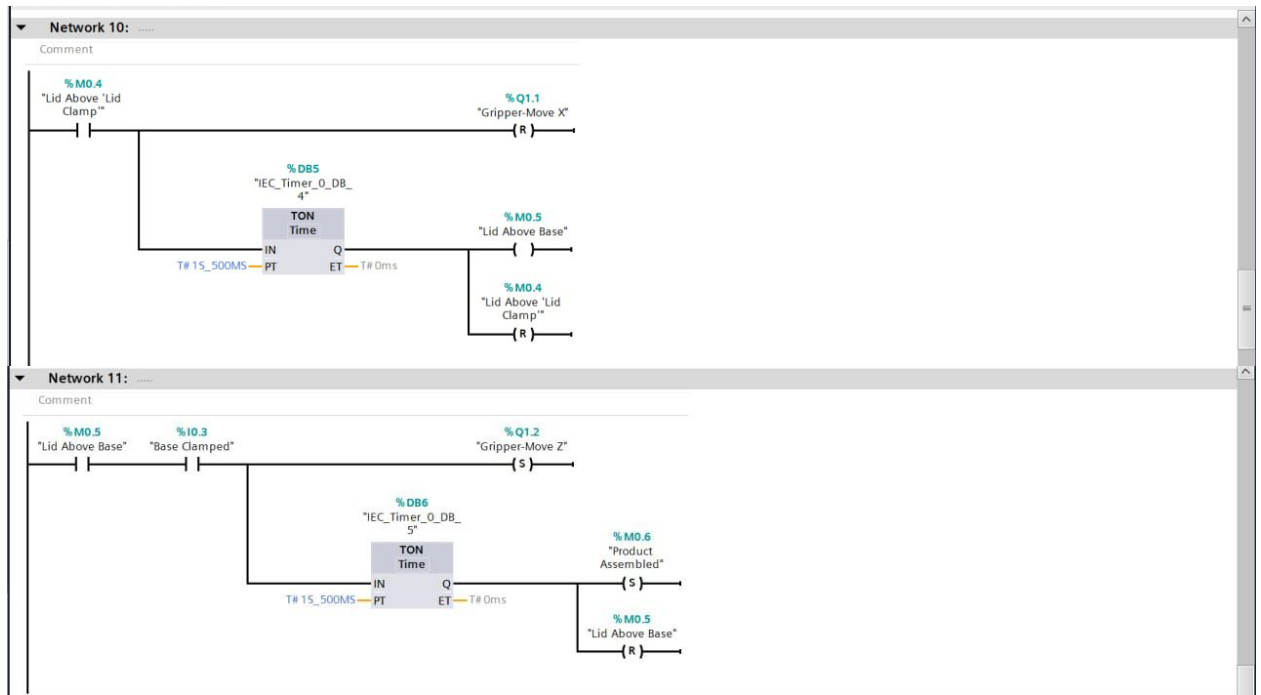


Figure 5 – Gripper Horizontal Retraction and Positional Sequencing

This logic initiates the reset of the gripper's X-axis movement once the lid is confirmed clear of the clamp, using a 1.5-second timer to sequence the transition of memory bits from "Lid Above Lid Clamp" to "Lid Above Base."

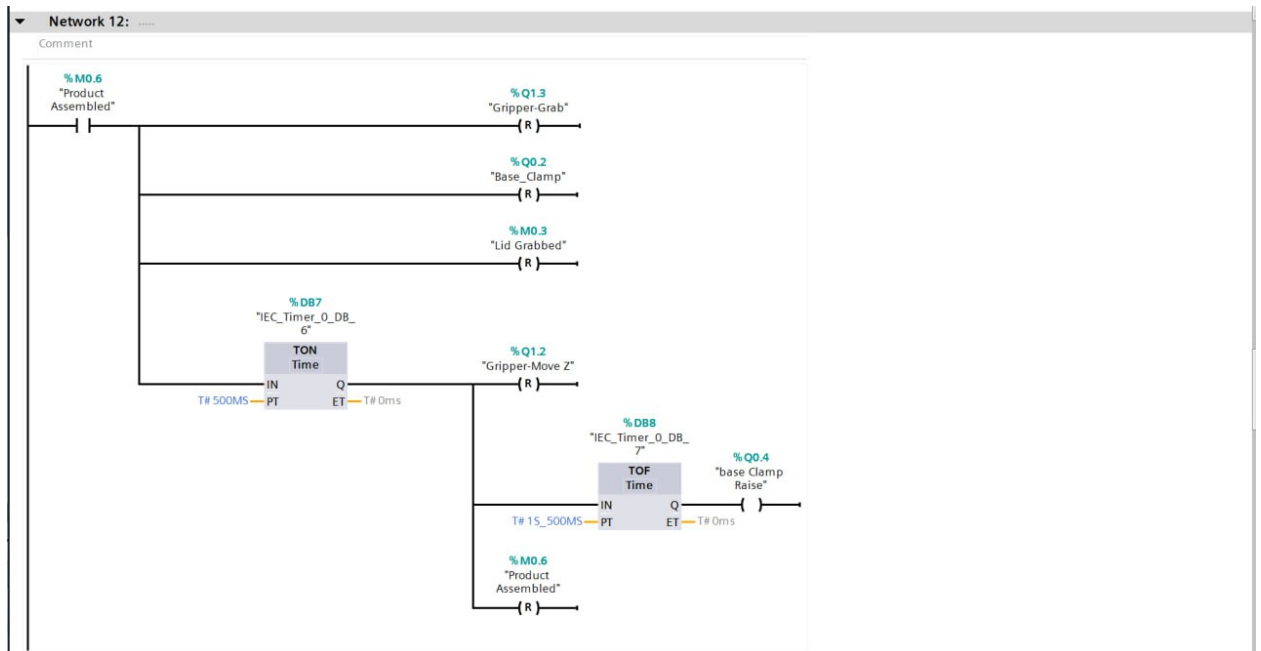


Figure 6 – Assembly Finalization and System Reset

Upon the activation of the "Product Assembled" memory bit, this logic resets the gripper's grab command, releases the base clamp, and utilizes a 500ms on-delay

timer to retract the Z-axis while triggering a 1.5-second pulse to raise the base clamp and clear all assembly status bits for the next cycle.

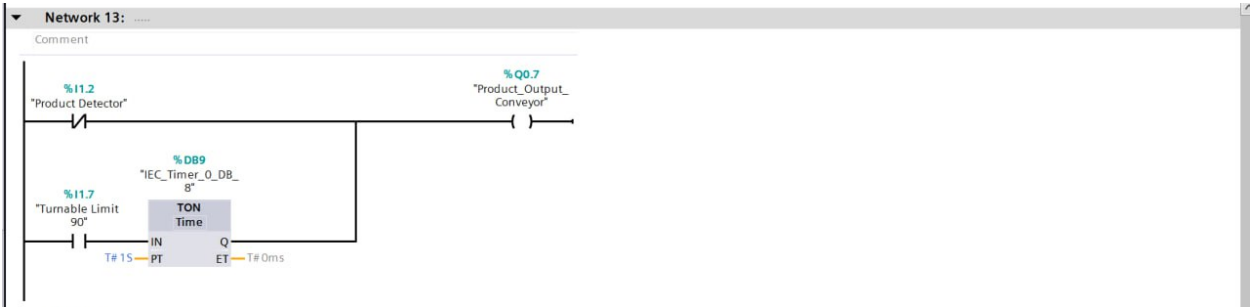


Figure 7 – Product Discharge and Conveyor Control

This logic manages the final output stage by activating the "Product Output Conveyor" coil whenever the product detector is clear or the turntable reaches its 90-degree limit after a 1-second delay, ensuring the completed module is transported out of the assembly area.

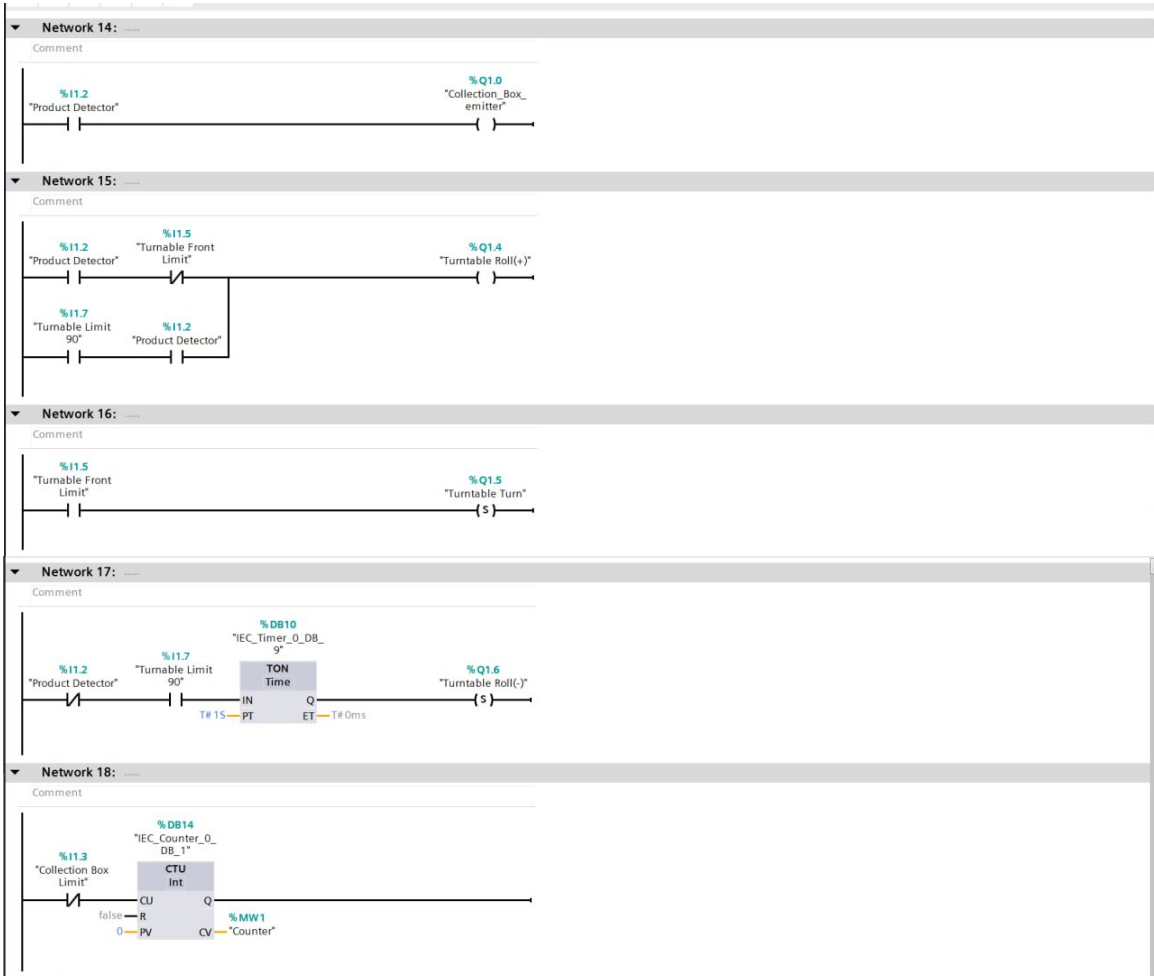


Figure 8 – Packaging Control

This logic coordinates the completion of the assembly cycle by resetting all gripper and clamping status bits, managing the synchronized movement of the product output conveyor and 90-degree turntable, and utilizing a counter to track finished units as they are emitted into the collection boxes.

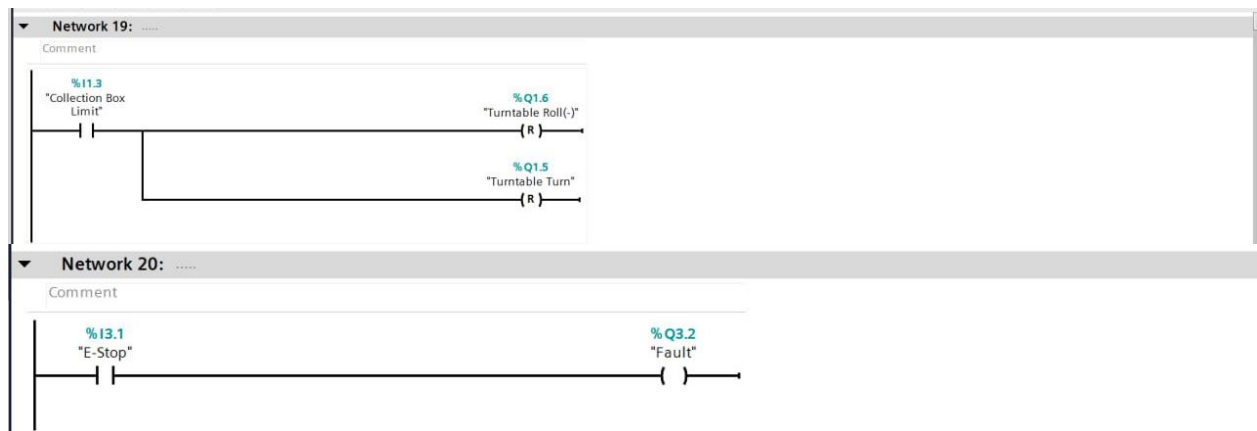
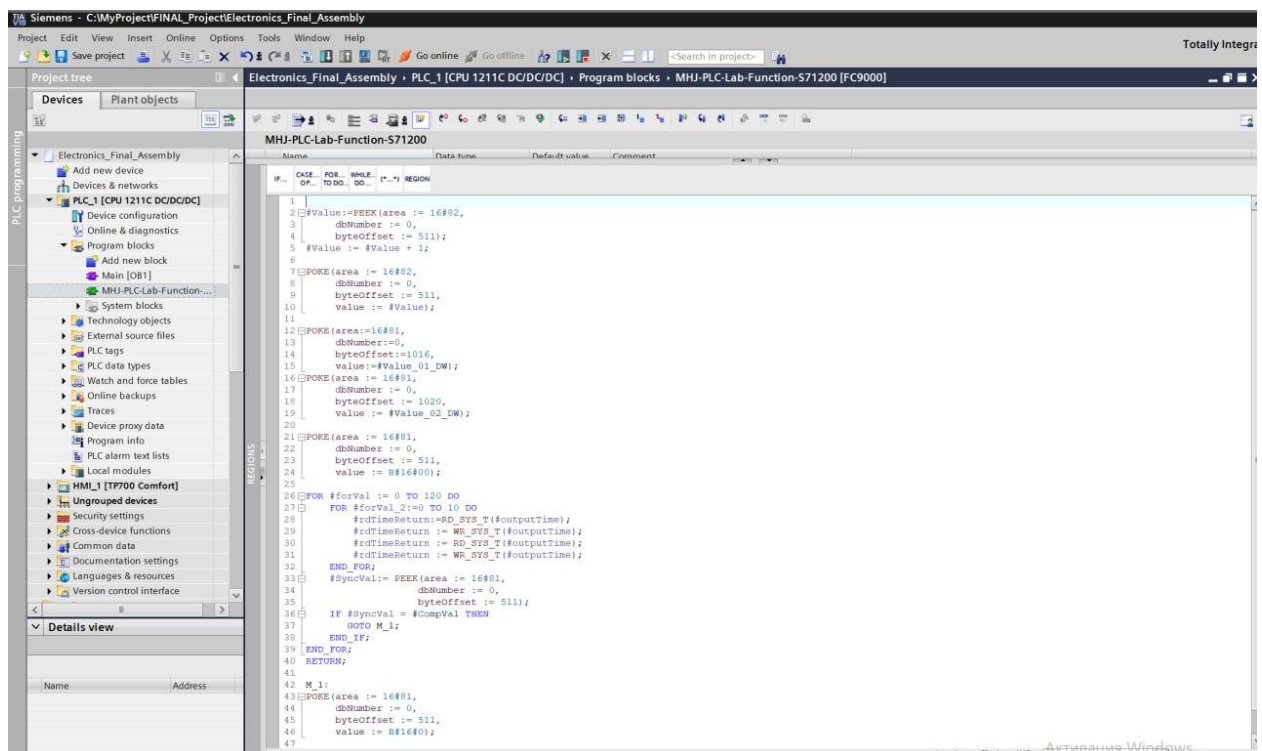


Figure 9 – Emergency stop and Fault

This logic initializes the system and enables material feed only when the Start button is pressed and no Stop or Emergency Stop is active. It ensures safe starts by preventing unintended operation. The "Start" signal is latched to maintain the run state after the button is released. Outputs activate both Base and Lid material emitters to begin the dual-stream process. This section is the foundation for all subsequent automation steps. Without it, the system cannot begin or respond safely to operator commands.



```

1
2 #Value:=PEEK(area := 16#82,
3   dbNumber := 0,
4   byteOffset := 511);
5 #Value := #Value + 1;
6
7 POKE(area := 16#82,
8   dbNumber := 0,
9   byteOffset := 511,
10  value := #Value);
11
12 POKE(area:=16#81,
13   dbNumber:=0,
14   byteOffset:=1016,
15   value:=#Value_01_DW);
16 POKE(area := 16#81,
17   dbNumber := 0,
18   byteOffset := 1020,
19   value := #Value_02_DW);
20
21 POKE(area := 16#81,
22   dbNumber := 0,
23   byteOffset := 511,
24   value := B#16#00);
25
26 FOR #forVal := 0 TO 120 DO
27   FOR #forVal_2:=0 TO 10 DO
28     #rdTimeReturn:=RD_SYS_T(#outputTime);
29     #rdTimeReturn := WR_SYS_T(#outputTime);
30     #rdTimeReturn := RD_SYS_T(#outputTime);
31     #rdTimeReturn := WR_SYS_T(#outputTime);
32   END_FOR;
33   #SyncVal:= PEEK(area := 16#81,
34     dbNumber := 0,
35     byteOffset := 511);
36   IF #SyncVal = #CompVal THEN
37     GOTO M_1;
38   END_IF;
39 END_FOR;
40 RETURN;
41
42 M_1:
43 POKE(area := 16#81,
44   dbNumber := 0,
45   byteOffset := 511,
46   value := B#16#0);
47

```

Figure 10 – Function Block using STL

This code ensures synchronization between the two parallel production streams in our automated assembly and packaging system. It reads and updates PLC memory values to keep the process steps aligned. Timing loops and conditional checks guarantee that robots, machining centers, and conveyors operate in sync. Without this logic, modules from different streams could arrive at the assembly station at the wrong time, causing errors.

Siemens - C:\MyProject\FINAL_Project\Electronics_Final_Assembly

Project Edit View Insert Online Options Tools Window Help

Save project Go online Go offline Search in projects

Totally Integr

Project tree Electronics_Final_Assembly • PLC_1 [CPU 1211C DC/DC] • PLC tags

Devices Plant objects

PLC programming

Electronics_Final_Assembly

Add new device

Devices & networks

PLC_1 [CPU 1211C DC/DC]

Device configuration

Online & diagnostics

Program blocks

Add new block

Main [OB1]

MHU-PLC-Lab-Function...

System blocks

Technology objects

External source files

PLC tags

Show all tags

Add new tag table

Standard-Variabletable...

PLC data types

Watch and force tables

Online backups

Traces

Device proxy data

Program info

PLC alarm text lists

Local modules

HMI_1 [TP700 Comfort]

Ungrouped devices

Security settings

Cross-device functions

Common data

Details view

Name

PLC tags	Name	Tag table	Data type	Address	Retain	Access...	Write...	Visibl...	Comment
1	Start	Standard-Variable...	Bool	%I0.0					
2	Stop	Standard-Variable...	Bool	%I0.1					
3	Base_Detector	Standard-Variable...	Bool	%I0.2					
4	Base_Clamped	Standard-Variable...	Bool	%I0.3					
5	Base_Clamp_Raised	Standard-Variable...	Bool	%I0.4					
6	Lid_Detector	Standard-Variable...	Bool	%I0.5					
7	Lid_Clamped	Standard-Variable...	Bool	%I0.6					
8	Gripper_Moving_X	Standard-Variable...	Bool	%I0.7					
9	Gripper_Moving_Z	Standard-Variable...	Bool	%I1.0					
10	Gripper-Touched lid	Standard-Variable...	Bool	%I1.1					
11	Product Detector	Standard-Variable...	Bool	%I1.2					
12	Collection Box Limit	Standard-Variable...	Bool	%I1.3					
13	Turnable Back Limit	Standard-Variable...	Bool	%I1.4					
14	Turnable Front Limit	Standard-Variable...	Bool	%I1.5					
15	Turnable Limit 90	Standard-Variable...	Bool	%I1.7					
16	Base_Material_emitter	Standard-Variable...	Bool	%Q3.0					
17	Lid_Material_Emitter	Standard-Variable...	Bool	%Q0.1					
18	Base_Clamp	Standard-Variable...	Bool	%Q0.2					
19	Lid_Output_Conveyor	Standard-Variable...	Bool	%Q0.3					
20	base_Clamp Raise	Standard-Variable...	Bool	%Q0.4					
21	Lid_Clamp	Standard-Variable...	Bool	%Q0.5					
22	Bottle Output Conveyor	Standard-Variable...	Bool	%Q0.6					
23	Product_Output_Conveyor	Standard-Variable...	Bool	%Q0.7					
24	Collection_Box_emitter	Standard-Variable...	Bool	%Q1.0					
25	Gripper-Move X	Standard-Variable...	Bool	%Q1.1					
26	Gripper-Move Z	Standard-Variable...	Bool	%Q1.2					
27	Gripper-Grab	Standard-Variable...	Bool	%Q1.3					
28	Turntable Roll(+)	Standard-Variable...	Bool	%Q1.4					
29	Turntable Turn	Standard-Variable...	Bool	%Q1.5					
30	Turntable Roll(-)	Standard-Variable...	Bool	%Q1.6					
31	Base Aligned	Standard-Variable...	Bool	%M0.1					
32	Lid Aligned	Standard-Variable...	Bool	%M0.2					
33	Lid Grabbed	Standard-Variable...	Bool	%M0.3					
34	Lid Above 'Lid Clamp'	Standard-Variable...	Bool	%M0.4					
35	Lid Above Base	Standard-Variable...	Bool	%M0.5					
36	Product Assembled	Standard-Variable...	Bool	%M0.6					
37	Emitter	Standard-Variable...	Bool	%M0.7					
38	Machining_Center_1_Progress	Standard-Variable...	Real	%ID30					

[illegible]

Figure 11 – PLC Tags

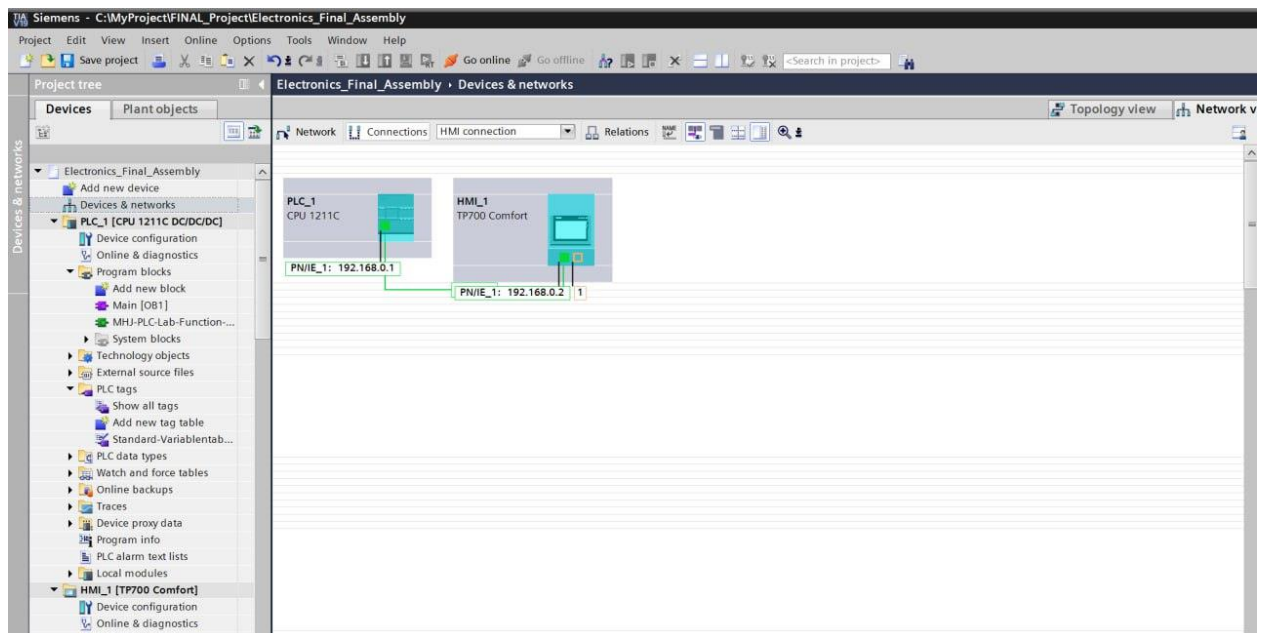


Figure 12 – PLC => HMI connection

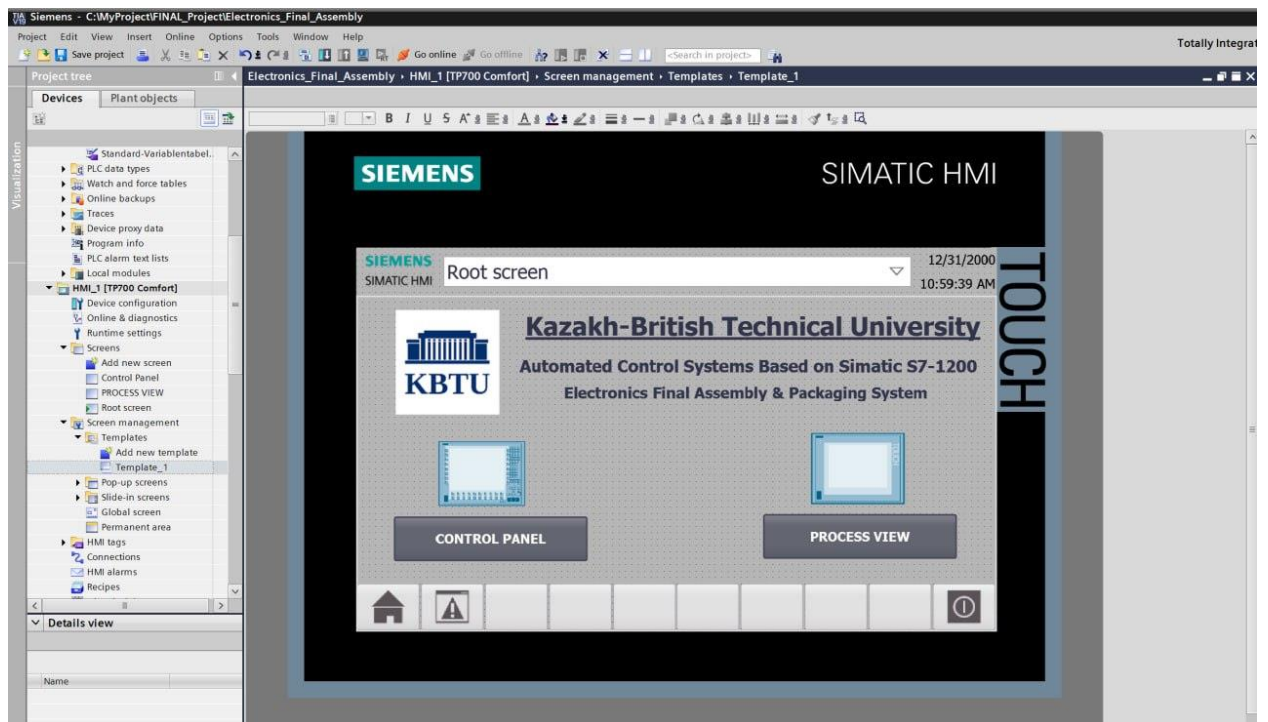


Figure 13 – HMI Main screen

Essential for operator interaction — allows switching between Automatic and Manual modes, starting/stopping the system, and triggering emergency stops. Displays real-time counters (box, product) and fault status. Enables manual control of robots and conveyors for setup or troubleshooting. Critical for safe, flexible operation without reprogramming PLC.

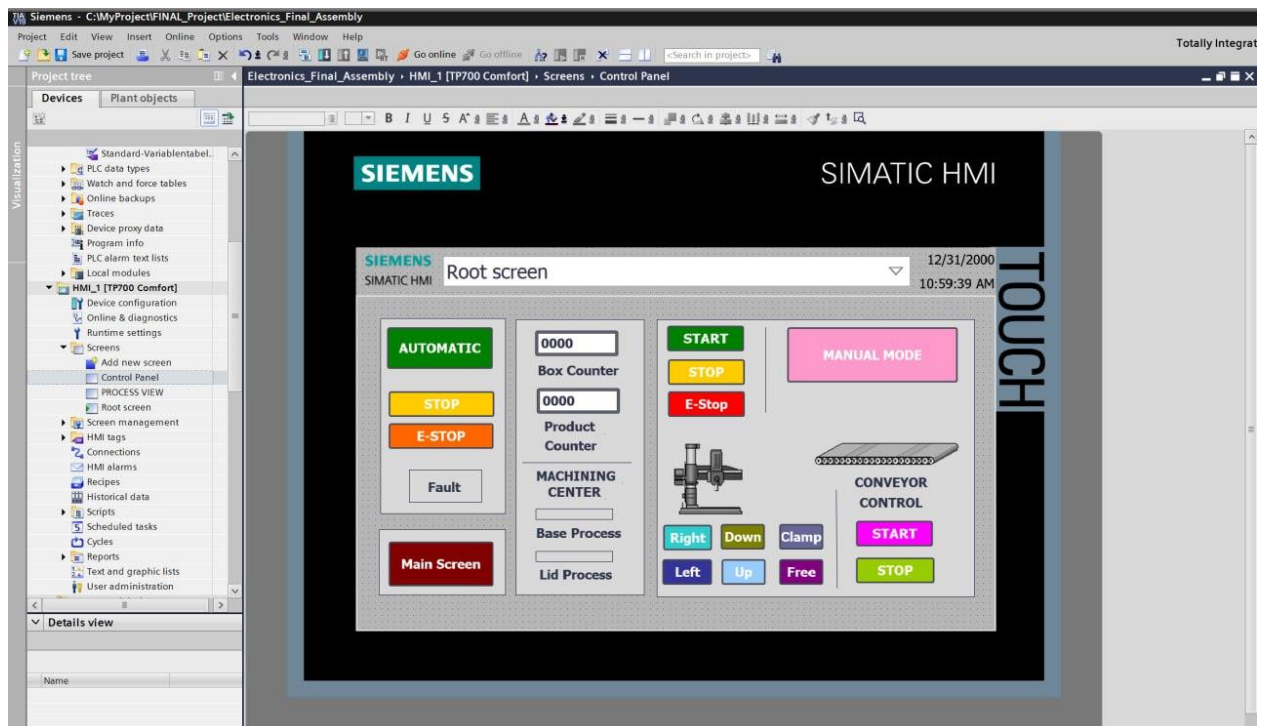


Figure 14 – HMI Control Panel

Provides visual overview of entire workflow — shows conveyor movement, robotic actions, machining, assembly, and packaging stages. Helps operators monitor system status and detect bottlenecks or errors immediately. Enhances traceability and operational awareness.

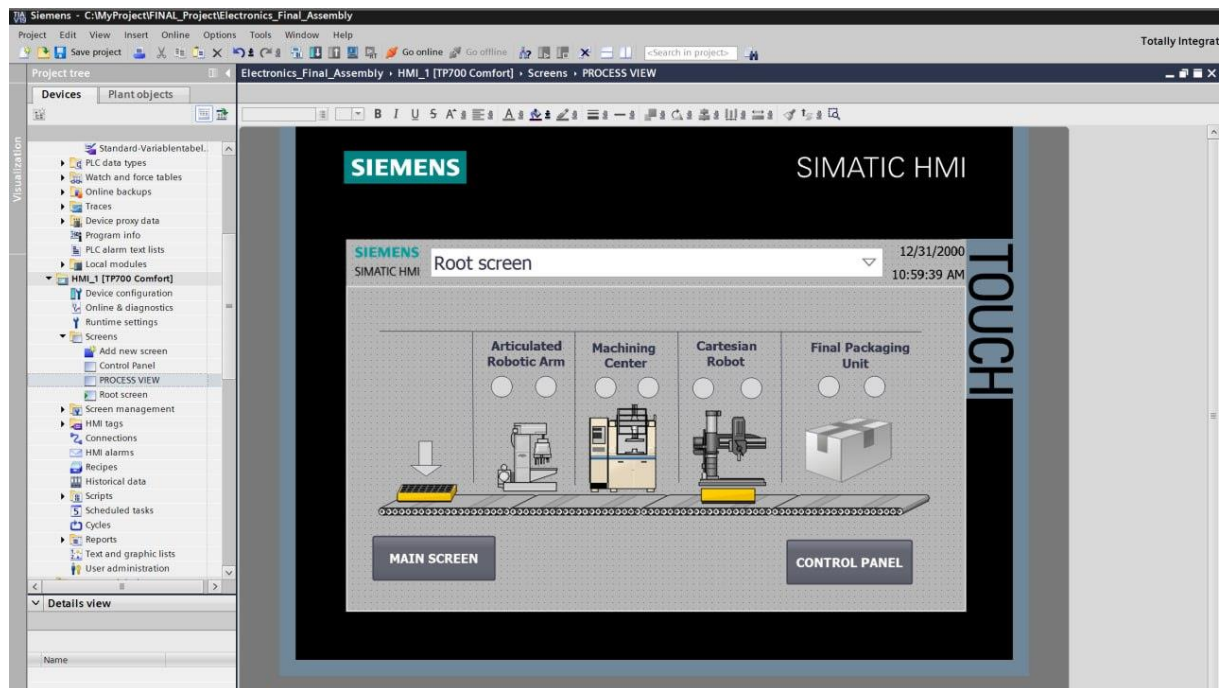


Figure 15 – HMI Process View

3. Technological Diagram

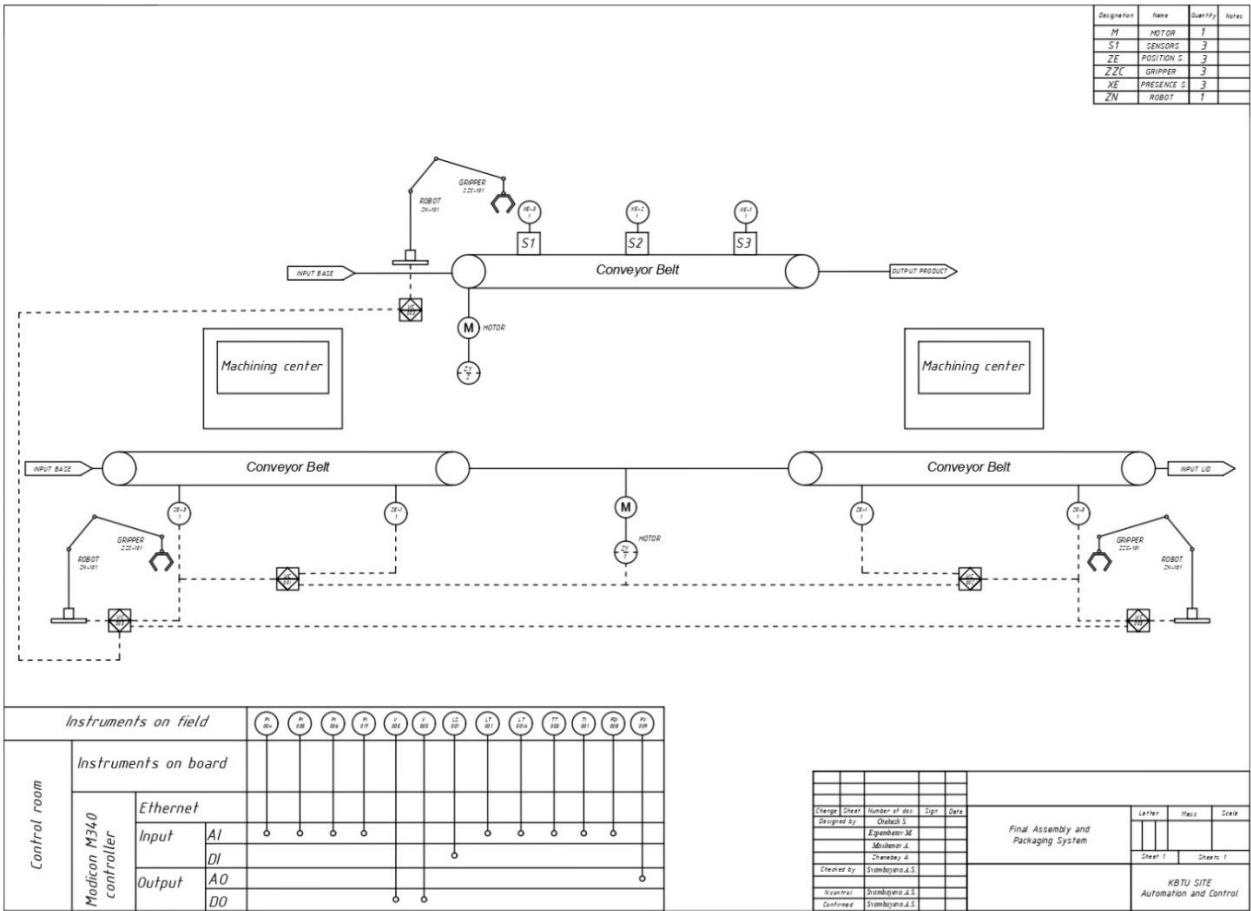


Figure 16 – Automation Scheme of process

Schematic diagram illustrates the physical architecture of the dual-stream production environment, mapping the field instruments and actuators to the PLC control logic. It details the spatial arrangement of the machining centers and conveyor belts where the material detection, clamping, and robotic pick-and-place operations—defined in Networks 3 through 9—physically occur. Finally, the diagram highlights the convergence of the base and lid streams into a single output path, representing the transition from the assembly logic in Network 12 to the final discharge and sorting control sequences.

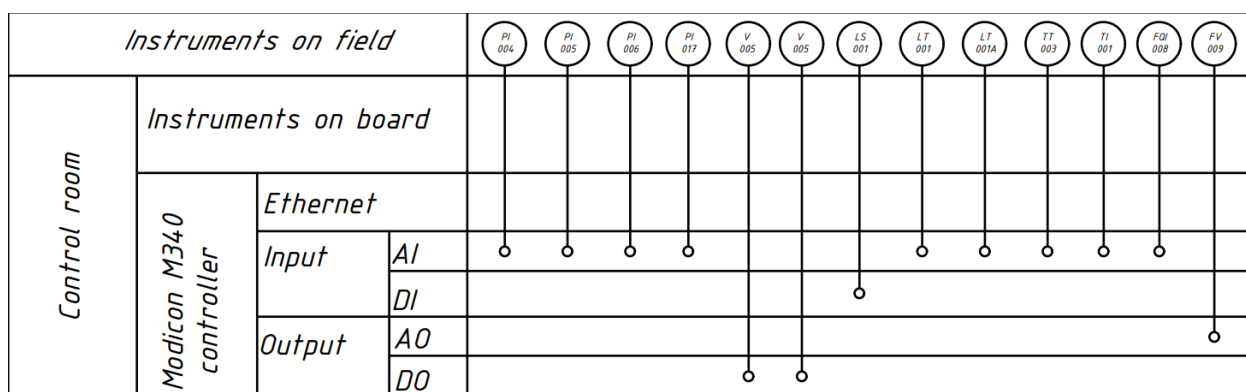


Figure 17 – Identification of all instrumentation

<i>Designation</i>	<i>Name</i>	<i>Quantity</i>	<i>Notes</i>
<i>M</i>	<i>MOTOR</i>	<i>1</i>	
<i>S1</i>	<i>SENSORS</i>	<i>3</i>	
<i>ZE</i>	<i>POSITION S.</i>	<i>3</i>	
<i>ZZC</i>	<i>GRIPPER</i>	<i>3</i>	
<i>XE</i>	<i>PRESENCE S.</i>	<i>3</i>	
<i>ZN</i>	<i>ROBOT</i>	<i>1</i>	

Figure 18 – Sensors and Actuators

CONCLUSION

This project focused on the design and simulation of a dual-stream automated final module assembly and packaging system using a Siemens S7-1200 PLC. The technological process was modeled as two parallel production lines, each consisting of conveyors, robotic manipulators, machining centers, and sensors. Modules enter the system via the conveyors, are picked and placed by robots into machining centers for secondary processing and then synchronized at a central assembly station to form a single integrated unit. The final product is subsequently transported to a packaging station, where it is enclosed, sealed, and prepared for output.

The technological scheme of the system includes logical flow diagrams representing material and control flows, with process variables such as conveyor speed, module presence, robot positions, machining time, and packaging status. Functional block representations were developed for each process unit—Conveyor FB, Robot FB, Machining FB, Assembly FB, and Packaging FB—to structure the PLC program modularly. Synchronization logic ensures both streams are coordinated before assembly, and interlocks maintain safe operation.

An HMI was integrated into the system for real-time monitoring, manual control, mode selection, and alarm handling. The software-based simulation of the process verified correct operation of all units under both automatic and manual modes. This project demonstrates how logical and functional technological schemes, combined with modular PLC programming and HMI supervision, enable precise, coordinated, and reliable control of complex discrete manufacturing systems.

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